hwsynth: Natural Synthetic Handwriting in 153 Unicode Scripts from Writer, Movement and Tool Models

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Abstract

Document understanding systems need handwriting in every script they meet, yet learned handwriting generators exist for a handful of scripts and need large per-script training sets, while rendering handwriting fonts looks mechanical: every *a* is the same *a*. We present `hwsynth`, a training-free generator that writes any text in 153 of the 175 Unicode scripts. Letters are traced from handwriting fonts (Noto where none exists) into pen paths; a writer model with 44 genes rewrites them; a movement model re-synthesises every stroke as overlapping lognormal submovements and trades shape fidelity against jerk energy by an effort parameter; a tool model with 17 writing instruments deposits ink by nib shape, ink rheology, speed and pressure. Twenty-four handwriting types, grounded in the discriminating elements of forensic handwriting examination, can each be written in ten situational styles and with either hand. A quality-diversity search (MAP-Elites) over measured slant and connectedness, scored against real scanned handwriting with document-disjoint fit and test sets and penalised for baseline instability beyond that of real handwriting, fills a bank of realistic individuals per type while keeping the types apart in two writer-identification spaces. On held-out real handwriting the searched bank reaches a Fréchet embedding distance of 0.158 against 0.433 for font rendering and 0.065 between two halves of the real data, and its 24 types are identified with 58 % top-1 accuracy (chance 4 %).

1 Introduction

Handwriting is everywhere in the documents that machines are asked to read: filled forms, letters, notes, prescriptions, archives. Training and testing text recognition, layout analysis and document dewarping on handwriting needs large amounts of it, with exact labels, in every script the system will meet. Collecting and annotating real handwriting at that scale is slow and, for most of the world's scripts, practically impossible.

Synthetic handwriting is the usual answer, and the field has largely converged on learned generators: recurrent networks for pen trajectories [10], GANs [8, 15], transformers [2, 23] and, most recently, diffusion models [21, 22]; see [5] for a survey. They produce convincing samples *for the scripts they were trained on*, almost always Latin and occasionally Arabic, Chinese or a few Indic scripts, because they need thousands of labelled lines per script and style. For the other scripts of Unicode, practitioners fall back to rendering handwriting fonts with random affine jitter. The result is recognisably mechanical: identical glyph instances, perfectly continuous strokes, uniform ink, straight baselines.

This paper asks whether a generator built from explicit models of *who* writes, *how the hand moves* and *what tool touches the paper* can close that gap without any trained network. The challenge is that each model must be faithful enough to remove the mechanical look, and their parameters must be set so that the output is both realistic and diverse, which no hand-tuning achieves across dozens of interacting parameters.

Contributions.

- A script-agnostic letter source: centrelines traced from handwriting fonts and Noto fallbacks, drawn per grapheme cluster or per shaped word, covering 153 Unicode scripts (§3.1).
- A writer model separating *identity* (forensic discriminating elements [14]) from *situation*, with 24 handwriting types × 10 styles × 2 hands (§3.2).
- A movement model combining lognormal submovements [24] with a minimum-jerk [7] trade-off controlled by effort, which orders the styles from medical scrawl to calligraphy (§3.3).
- A tool and ink model for 17 instruments after the ink-deposition classes of [9] (§3.5).
- A leakage-controlled evaluation and a quality-diversity search that scores realism against real handwriting and separation with image and pen-path writer-identification features [3, 17] (§3.6).

The code is released under CC BY-NC 4.0 (§6).

2 Related Work

Learned handwriting generation. Graves [10] generated online pen trajectories with an LSTM and mixture density outputs. Haines et al. [11] synthesised a specific person's handwriting by assembling and blending their glyphs. ScrabbleGAN [8] and GANwriting [15] generate offline word images conditioned on content and style; Handwriting Transformers [2] and VATr [23] condition on a few style samples; latent diffusion models [21, 22] currently give the most faithful styles. All require labelled handwriting in the target script and style, and none covers more than a few scripts.

Perturbation and font based synthesis. Before deep generators, synthetic training data came from geometric perturbation of real lines [31] and from fonts. Font rendering scales to any script with a font, but repeats glyphs exactly; our letter source keeps the coverage of fonts and removes the repetition by drawing every glyph as a fresh movement.

Models of handwriting movement. The Kinematic Theory [24, 25] represents a stroke as a sum of lognormal velocity impulses; its Sigma-Lognormal model reconstructs real trajectories [6] and has been used to design trajectories [1]. Minimum-jerk planning [7] explains the smoothness of point-to-point arm movements. We use both: lognormal impulses for timing and corner cutting, jerk energy for the effort trade-off.

Individuality of handwriting. Forensic examiners compare a set of discriminating elements: allograph design, connections, proportions, slant, spacing, baseline, commencements and terminations, line quality and natural variation [14]. Srihari et al. [29] quantified individuality statistically. Automatic writer identification uses contour-direction and contour-hinge distributions [3, 26] or path signatures [17], grounded in rough path theory [4, 19]. We use these features to measure and enforce that our types are distinguishable.

Ink and pencil. Franke and Rose [9] related pen force to ink deposition for solid, viscous and fluid inks; Sousa and Buchanan [28] modelled graphite on paper. Our tool model follows these classes at the cost of a renderer, not a physical simulation.

Evaluation and search. We report Fréchet distances [13] and k-NN precision and recall [16] on ImageNet ResNet18 [12] embeddings and on stroke features. The writer bank is filled by MAP-Elites [20], a quality-diversity algorithm that keeps the best solution in every cell of a behaviour grid, related to novelty search [18].

text → scripts, bidi order, units → **letters** (font centrelines) → **writer** (identity + style genes: shape, layout, joins) → **hand** (lognormal sub-movements, minimum jerk, wrist pivots, tremor) → **tool** (width, density, texture) → **page** (paper, ruling, smudges, word boxes)

Figure 1: The generation pipeline. No stage is learned; the writer bank (§3.6) only selects genes.

3 Method

Figure 1 summarises the pipeline. Text is split into words and drawing units; each unit's glyph centreline is transformed by the writer, the strokes of a line are moved by the hand, and the tool deposits ink.

3.1 Letters in every script

Each character is assigned its Unicode script [30]. Scripts with independent letter forms (Latin, Cyrillic, Greek, Hebrew, Han, Hangul, …) are drawn per grapheme cluster, so every letter gets its own jitter; all others are shaped by HarfBuzz/Raqm and drawn per word, so joining and conjuncts come from the font. Right-to-left runs are reversed while digits and Latin runs keep their order. A writer uses its Latin handwriting font for any script that font covers, otherwise a handwriting font of the script (59 fonts with class labels cursive, print, brush), otherwise a Noto Sans font (158). A glyph is rasterised at 160 px, skeletonised, and traced into a graph whose free spurs are pruned and whose edges continue straight through junctions (smallest turning angle), giving ordered pen strokes. Glyph size is normalised per script by a reference letter (e.g. Han is written at twice the x-height). Of the 175 scripts in Unicode, 153 have at least one font covering 90 % of their core letters.

3.2 Writers: identity and situation

A writer is a vector of 44 genes in $[0, 1]$. Twenty-three are *identity* genes following the discriminating elements of [14]: size, lateral expansion, ascender and descender length, capital size, slant, letter and word spacing, line pitch, roundness and angularity (garland, arcade, angular and thread movement), joining, entry hooks and exit flicks, dot placement and form, pressure, the allograph set (font), the share of letters written in a second allograph, handedness, all-capital writing, zone flattening and circle-shaped dots. The others are *situational*: jitter of size, slant, rotation, baseline and spacing (correlated between neighbouring letters, AR(1) with $\rho = 0.6$), speed, effort, decay towards word ends, slant drift, baseline wander, tremor, cross-outs, retracing and mixed capitals.

A *type* fixes a centre for every identity gene (twenty-four types, Table 4 in the appendix); an individual draws each centred gene within ±12 % of its range. A *style*

remaps situational genes into style ranges, keeping each writer's relative position, and shifts a few identity genes (e.g. hurried writers join more and slant further). Left-handed writers (hand gene > 0.9, about 10 %) pull horizontal strokes from right to left, draw some loops clockwise, write more upright, hold broad nibs steeper and smudge wet ink.

Natural variation comes from four sources: a second allograph for a fixed share of letters, AR(1) letter jitter, stroke ends that overshoot or stop short, and per-stroke ink-flow drift. Rare events (cross-outs of a misspelt word, retraced strokes) are drawn from skewed priors.

3.3 Hand movement

Each stroke is reduced to virtual targets (Douglas–Peucker turning points). Segment j becomes a sub-movement with velocity $\mathbf{D}_j \Lambda(t; t_{0j}, \mu_j, \sigma_j)$, where Λ is the lognormal impulse response of the Kinematic Theory [24]; start times are spaced by $2\,o\,e^{\mu-\sigma^2}$, twice the overlap ratio o times the time of peak speed (isochrony). Fluent writers have small o and cut corners; discrete-stroke scripts (Han, kana, Hangul, Devanagari) get larger o. The integrated path is then the minimiser of

$$\min_{\mathbf{x}} \ \lambda(\epsilon)\,\|\mathbf{x}-\mathbf{p}\|^2+\|D_3\mathbf{x}\|^2, \qquad \log_{10}\lambda = -6+6.5\,\epsilon, \quad (1)$$

with D_3 the third difference (jerk) and end points held: a writer with low effort ϵ follows the minimum-jerk path and lets the letter shape go, a writer with high effort keeps the shape and pays in jerk energy. Effort is a situational gene whose style ranges increase from medical (0–0.12) through hurried, professional, casual, educated and formal to accurate and calligraphic (0.9–1). The speed of the synthesised movement drives the tool. Lines follow wrist-pivot arcs that reset at hand shifts every 25–60 mm, plus slow random baseline wander.

3.4 Mathematics, chemistry and figures

Inline `$...$` segments use a small TeX-like syntax: scripts, fractions, roots, Greek letters and symbols, drawn arrows, `\ce{...}` chemical equations (counts become subscripts, charges superscripts, reaction and equilibrium arrows) and `\benzene`. They are laid out as boxes of pen strokes: glyph runs come from the writer, rules, radicals, arrows and rings are drawn as strokes, and the writer's hand then moves over the whole expression like a word. Figures (circles, ellipses, triangles, rectangles, arrows, stars, ticks, crosses, axes with a curve, underlines) are planned as freehand paths that do not quite close, overshoot at corners and bow slightly, and are then moved and inked by the same models.

3.5 Tools and ink

Seventeen tools combine a nib shape (round; broad edge with thick–thin contrast $w + e|\sin(\phi+\alpha)|$ for direction ϕ and nib angle α; brush with width growing as pressure$^{1.5}$ and tapered ends) with an ink class [9]: *solid* (graphite: density follows force, paper grain), *viscous* (ballpoint paste: density falls with speed, skips on fast strokes, a blob where the pen lands) and *fluid* (gel, rollerball, fineliner, fountain, markers: near-constant density, pooling at slow points, bleeding halo, darker overlaps). Pressure combines the writer's push, its variation, extra force on downstrokes, and lift-off ramps. Strokes are drawn as dense anti-aliased discs at 3× supersampling and combined with tool-specific overlap.

3.6 Writer bank: quality-diversity search

Real references. Text lines are cut from 980 scanned handwritten documents, cleaned (no rulings, staves or bars; contrast; ink share) and kept when most of their five nearest labelled lines are handwriting. The documents are split into a *fit* set (2 876 lines) used by the search and a *test* set (1 181 lines) used only for metrics; documents whose lines were used to label the handwriting filter are never test documents.

Priors. Genes a type leaves free are not drawn uniformly: noise genes are drawn as u^k (k = 2–4), concentrated near zero, and signed drifts near their centre. Uniform priors gave the median writer half of the maximum jitter on every noise channel at once, which looks unstable and childish; most adults vary narrowly to moderately. Fonts with a childish design are reserved for the child type and the uneducated style.

Stability from real handwriting. Baseline instability is the standard deviation of word ink centres around their fitted line, in x-heights (estimated from the dense core band of the row profile). On 1 013 lines from fit documents it runs from about 0.13 for steady to 0.35 for wavy clean lines; higher values belong to extraction errors (tables, staves, two lines). Generated and real lines are measured with the same function.

Search. Each type has a MAP-Elites archive [20] over two measured behaviour descriptors: the slant of near-vertical pen-path segments (7 bins) and pen lifts per letter (5 bins). An individual is rendered on four search texts, in the casual style and in a random other style. Its quality is

$$Q = \mathit{realism} - 4\max(0,\ \mathit{instability} - 0.15) - [\mathit{separation} \le 0], \quad (2)$$

where *realism* is the mean of two z-scores (cosine similarity of its line embeddings to their five nearest fit lines; minus the distance of its stroke features to their five nearest fit lines; both standardised by the same statistic of fit lines against lines of other documents) and *separation* is the distance to the nearest other type's centroid minus the distance to its own, in contour-direction/hinge

space [3] (χ^2) and in path-signature space [17], each relative to the own distance. New individuals are iso-line variations of two elites of the same type, kept within the type's range. Finally every elite is re-evaluated with two fresh seeds and banked when its realism is at least -0.3, it is nearer its own type than any other, and its instability stays below 0.35.

4 Experiments

Setup. Metrics use texts disjoint from the search texts and the held-out test documents. We compare: plain rendering of a handwriting font (*font*); writers with uniform random genes (*random*); type priors without search (*prior*); the searched bank (*bank*); and the bank without the pen model, without shape variation, and without letter joins. Embeddings are ResNet18 (ImageNet) [12]; legibility is the character error rate of Tesseract [27] on English lines.

4.1 Realism and diversity

Table 1 compares the systems on the held-out test documents. Lower Fréchet distances (FD) and higher precision and recall mean closer to real handwriting; the first row is the reference level of real handwriting against other real handwriting.

4.2 Styles and effort

Table 2 lists the bank per style. Movement energy (jerk per unit length of the planned pen path) and turning entropy rise with the effort a style spends on letter shapes, from medical scrawl to calligraphy; legibility follows.

4.3 Distinguishable types, individuals and styles

Table 3 reports top-1 nearest-neighbour identification on held-out texts: a type must be recognised from *other* individuals of the type, an individual from its *other* texts, a style from *other* individuals.

Analysis. The searched bank closes three quarters of the gap between font rendering and real handwriting in embedding FD (0.158 against 0.433 for fonts and 0.065 between real documents), with precision 0.69 against 0.34 for fonts. The scan model is the largest single contributor: without it FD rises to 0.249 and recall falls from 6.4 % to 1.6 %, although the realism z-score, which saturates and treats scanner noise as atypical, moves the other way (0.48 against 0.40; Figure 2). Recall remains low: real scans span archives, coloured paper and damage that the model only partly covers. The search adds realism (0.40 against 0.28 for the type priors) and precision (0.69 against 0.59) while keeping instability inside the real range (Figure 3). Removing shape variation or letter joins lowers realism and precision; removing the pen model lowers precision and raises FD. Movement energy rises strictly with the effort of the styles, from medical (1.7×10^{-7}) through hurried, professional, casual, educated and formal to calligraphic (3.9×10^{-5}), and turning entropy rises with it. Accurate writing is the most legible style for Tesseract (CER 0.17) and medical writing the least (0.66). Generation with the scan model takes about 0.3 s per line on one CPU core.

5 Discussion and Limitations

The realism measure is a proxy: ImageNet embeddings and 22 stroke statistics see texture and layout, not the correctness of letterforms in a particular script, and the reference lines are mostly Latin. No trained writer identification network was available, so separation uses classical features; the deep writer embeddings of recent work would be a stronger test. Scripts without a handwriting font inherit Noto's letterforms, which our writer and movement models make handwritten in execution but not in allograph design. Fitting individuals to real documents by analysis-by-synthesis would ground the bank in real writers.

6 Conclusion

Explicit models of the writer, the hand and the tool, set by a leakage-controlled quality-diversity search, produce natural, diverse handwriting in 153 scripts without any trained generator. Every layer is inspectable: one can ask for a left-handed elderly writer, in a hurry, with a pencil. Natural extensions are handwriting fonts for the remaining scripts, page-level phenomena (margins, insertions, marginalia), learned discriminators as realism judges, and fitting the bank to real writers. Code, examples and this paper: `https://github.com/pavel9860`.

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system	realism	FD emb.	FD feat.	precision	recall	diversity	writer id	CER	ms/line
real (other documents)	–	0.065	3.70	0.746	0.658	–	–	–	–
font rendering	-0.428	0.433	28.11	0.338	0.000	0.0941	0.106	0.048	10
random writers	0.242	0.180	18.36	0.519	0.070	0.1702	0.350	0.587	536
type priors	0.280	0.160	17.13	0.594	0.082	0.1755	0.338	0.451	369
searched bank	0.402	0.158	16.45	0.688	0.064	0.1679	0.338	0.390	309
bank, no scan model	0.480	0.249	18.70	0.800	0.016	0.1202	0.350	0.385	461
bank, no pen model	0.403	0.171	16.38	0.662	0.047	0.1618	0.287	0.366	307
bank, no shape variation	0.376	0.169	16.12	0.600	0.057	0.1702	0.325	0.396	445
bank, no joins	0.346	0.157	16.32	0.606	0.065	0.1729	0.287	0.328	308

Table 1: Systems against held-out real handwriting (casual style, texts disjoint from the search texts).

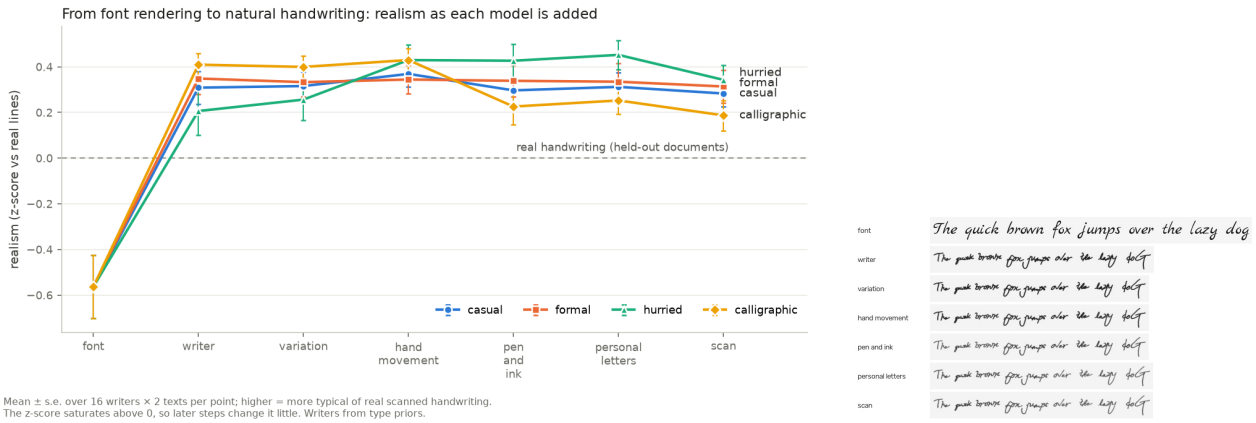


Figure 2: From font rendering to natural handwriting. Left: realism on held-out real lines as each model is added (mean ± s.e., 16 writers × 2 texts per point). The measure saturates once the output looks like handwriting, so later stages change it little. Pen and ink lowers the calligraphic style, whose broad-nib contrast is rare in the mostly ballpoint references. The scan look lowers it slightly too. Right: one writer at each stage.

style	realism	FD emb.	energy	entropy	CER
medical	0.239	0.201	1.7e-07	0.908	0.662
hurried	0.376	0.174	3.2e-07	0.931	0.544
professional	0.347	0.195	7.6e-07	0.950	0.415
casual	0.402	0.158	1.5e-06	0.974	0.390
unsupported	0.457	0.173	1.5e-06	0.966	0.394
educated	0.350	0.175	3.7e-06	0.996	0.407
uneducated	0.312	0.197	1.8e-06	0.968	0.249
formal	0.331	0.176	6.4e-06	1.000	0.348
accurate	0.423	0.203	9e-06	0.998	0.172
calligraphic	0.194	0.191	3.9e-05	1.203	0.548

Table 2: Searched bank per style, ordered by effort.

writers	features	type	individual	style
prior	contour hinge	0.312	0.384	0.442
prior	path signature	0.292	0.542	0.484
bank	contour hinge	0.520	0.342	0.510
bank	path signature	0.582	0.454	0.604
chance		0.042	0.0104	0.200

Table 3: Identification accuracy (24 types, 5 styles).

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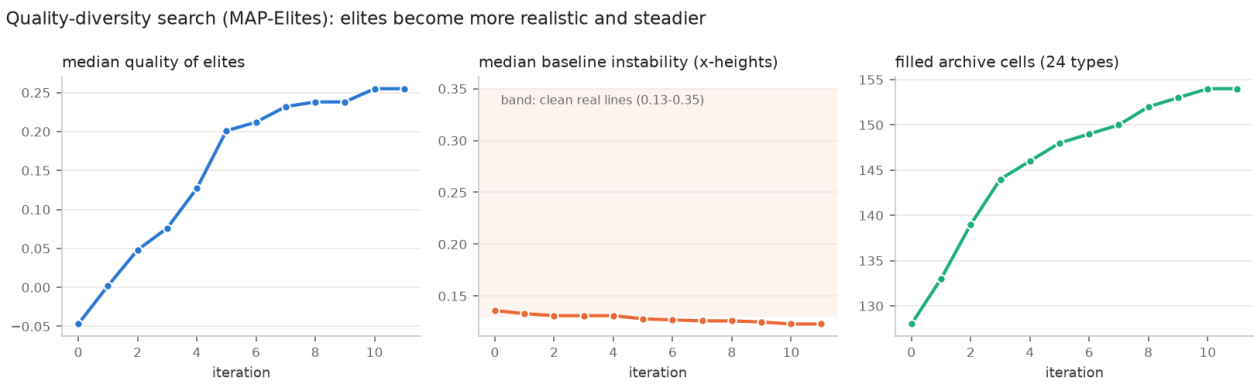


Figure 3: Quality-diversity search over all 24 types: median elite quality rises from −0.05 to 0.26, median baseline instability falls inside the range of clean real lines, and the archive fills 154 cells.

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A Handwriting types and styles

type	fonts	description
school cursive	cursive	taught joined script: loops, right slant, garlands
round print	print	rounded upright print, open letters
compact print	print	small neat narrow print
block capitals	print	everything in capitals, drafting style
backhand	any	left-leaning script
angular cursive	cursive	sharp, narrow, strongly slanted joined script
thready	cursive	fast thread connections, letters dissolve
wide garland	cursive	wide open letters, generous spacing
narrow tall	any	narrow letters, long ascenders and descenders, tight spacing
micro	any	very small writing
expansive	any	large writing, big capitals, wide spacing
arcade	cursive	rounded tops, slow upright joined script
print script	any	print and script mixed
elderly	cursive	slow tremulous joined script
child	print	large print, uneven baseline, heavy pressure
left hooked	any	entry hooks, slight left slant, pressure on downstrokes
copperplate	cursive	steep slant, long loops, ornate capitals
sloppy fast	cursive	fluent but careless
ruled neat	any	very regular, even and aligned
rounded neat	print	round, upright, regular and open: a rounded neat hand
bubbly print	print	bubbly rounded print with circle dots
looped cursive	cursive	rounded joined script with full ascender and descender loops
soft upright	any	soft upright hand, rounded joins, even size
artistic	cursive	embellished capitals, exaggerated loops and terminals
style	**tools**	**description**
casual	all	everyday notes: as the person writes
formal	fountain fine, fountain medium, ballpoint	letters, applications: careful and even
accurate	ballpoint, gel, fineliner	forms and labels: separated careful letters
calligraphic	broad nib, italic nib, brush pen	broad-nib pen calligraphy: slow, contrasted
professional	ballpoint, gel, rollerball	fast, fluent, consistent office notes
hurried	ballpoint, ballpoint fine, gel	written in a rush
medical	ballpoint, ballpoint fine, gel	prescription scrawl
educated	ballpoint, fountain fine, fountain medium	fluent practised writing
uneducated	ballpoint, pencil 2b, pencil hb	slow unpractised print, mixed capitals
unsupported	ballpoint, gel, pencil hb	on a sheet held in the hand, no support

Table 4: The 20 handwriting types and 10 styles; 17 tools.

B Examples

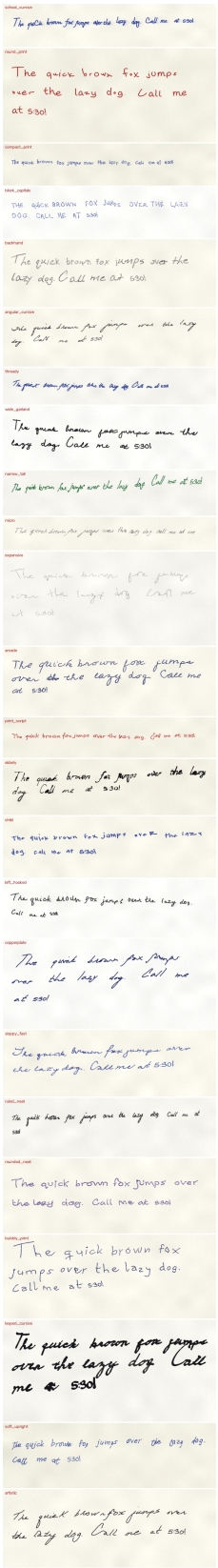


Figure 4: One individual of each of the 24 types (casual style).

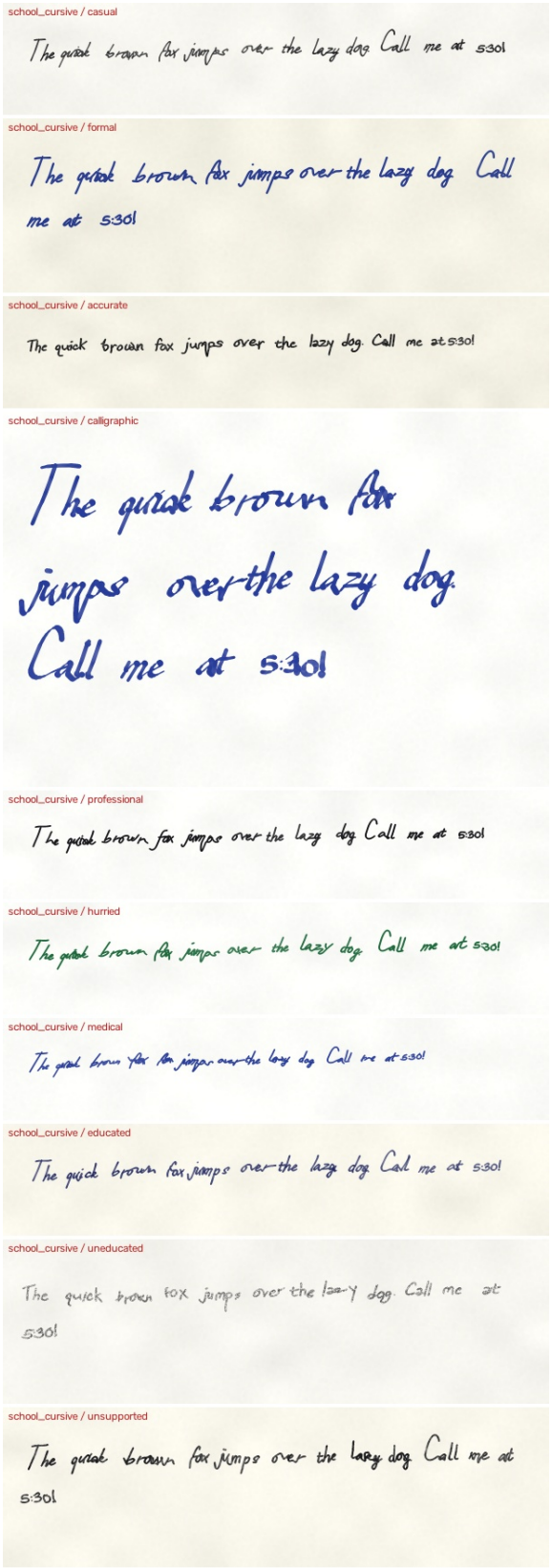


Figure 5: One school-cursive individual in the 10 styles.

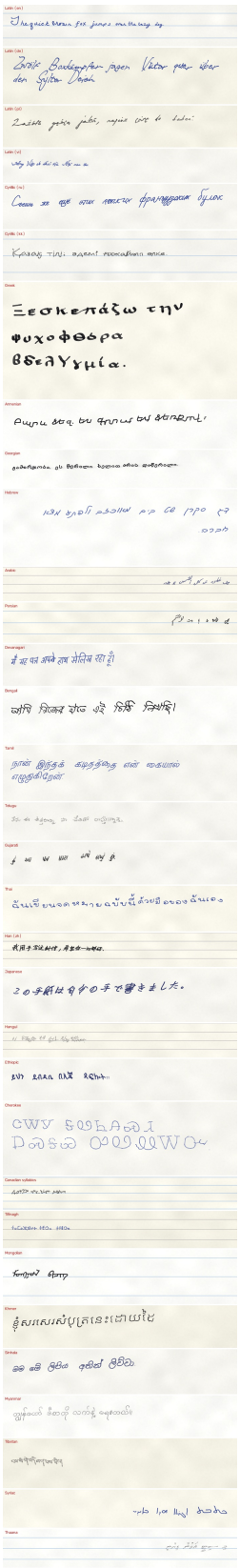


Figure 6: Random individuals writing 32 scripts.

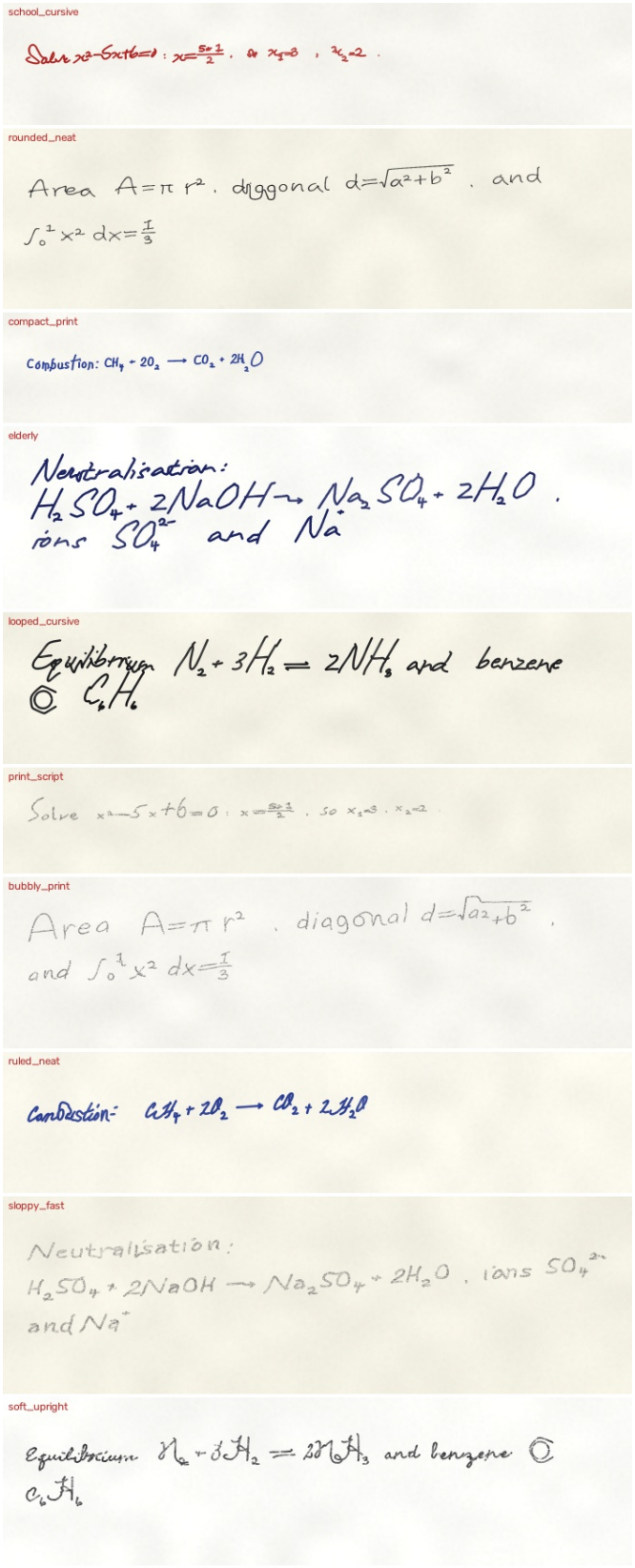


Figure 7: Inline mathematics and chemistry.

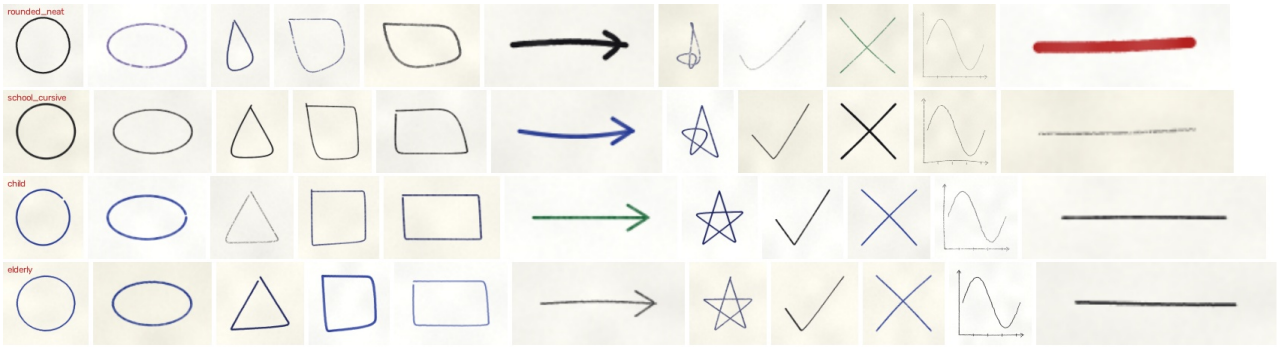


Figure 8: Hand-drawn figures by four types.



Figure 9: The 17 tools at light and strong push.

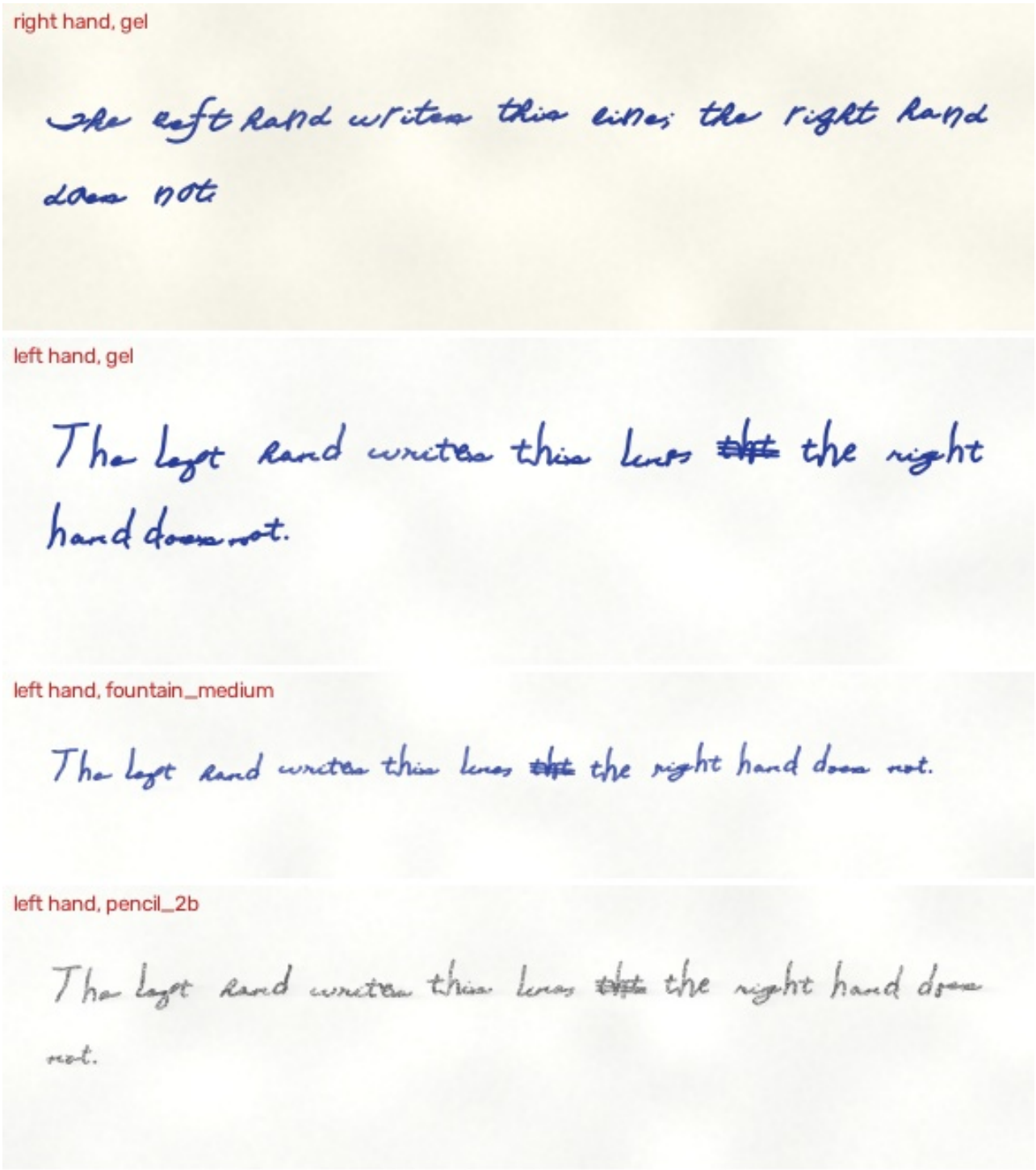


Figure 10: Right hand and left hand (gel, fountain pen, pencil): upright slant and smudges.